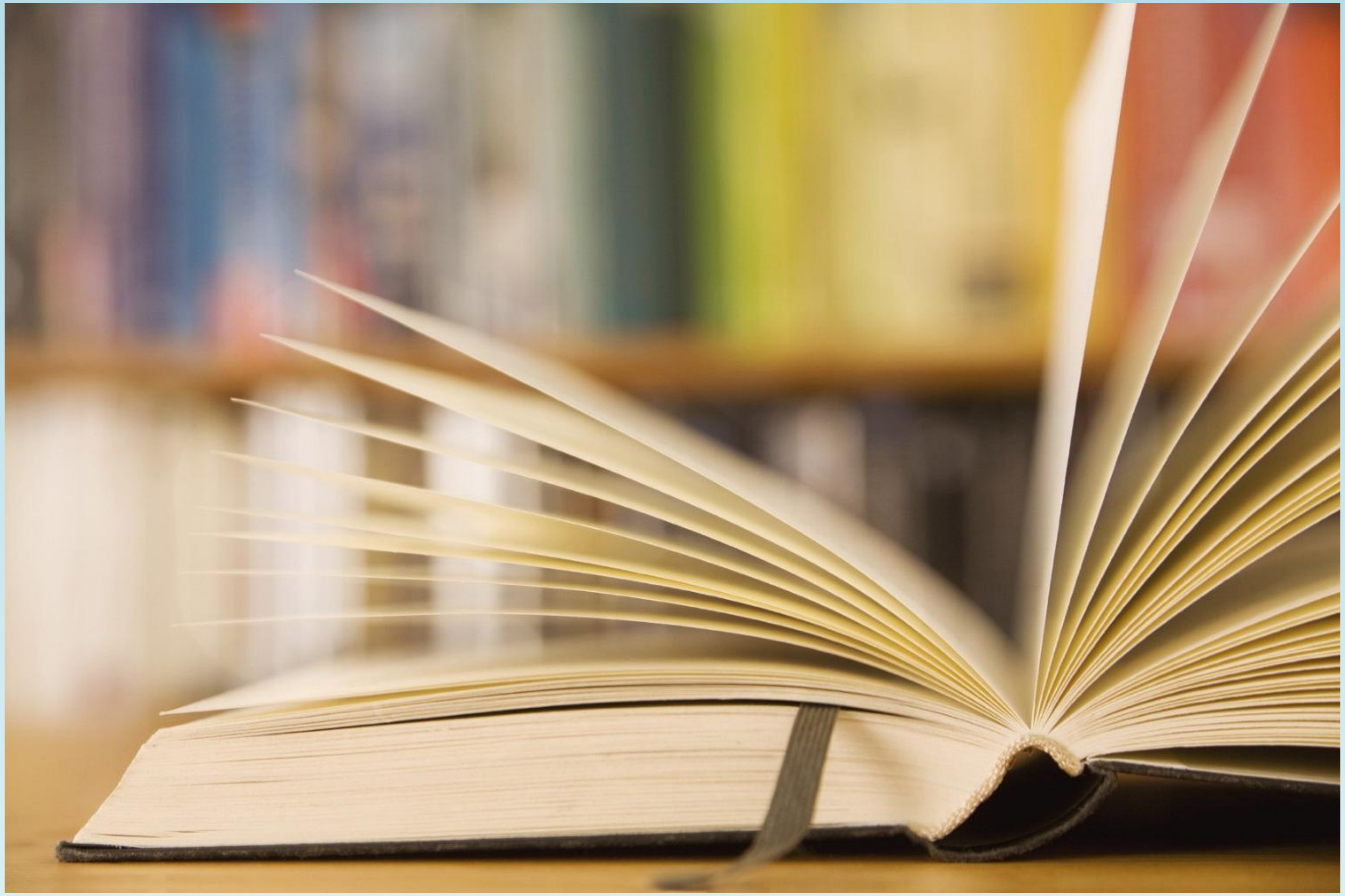


The Systems Analyst

BIS601

Systems Analyst

- Many projects abandoned because the analysts tried to build a wonderful system without clearly understand how the system would support the organization’s goals, improve business processes, and integrate with other IS to provide value.

Systems Analyst Skills

- Technical: existing technical environment, new system's technology foundation, an integrated technical solution
- Business: how IT can be applied to business situations and delivers real business value
- Analytical: solve problems at both the project and organizational level

Systems Analyst Skills (cont.)

- Interpersonal: communicate, give presentations, write report
- Management: manage people, pressure, risks
- Ethical: maintain confidence and trust

Systems Analyst Roles

- Systems analyst: IS issues surrounding the system – ways that IT can support and improve business processes
- Business analyst: business issues surrounding the system – identify business values that the system will create, develop ideas for improving business processes
- Requirement analyst: eliciting requirements from stakeholders

Systems Analyst Roles (cont.)

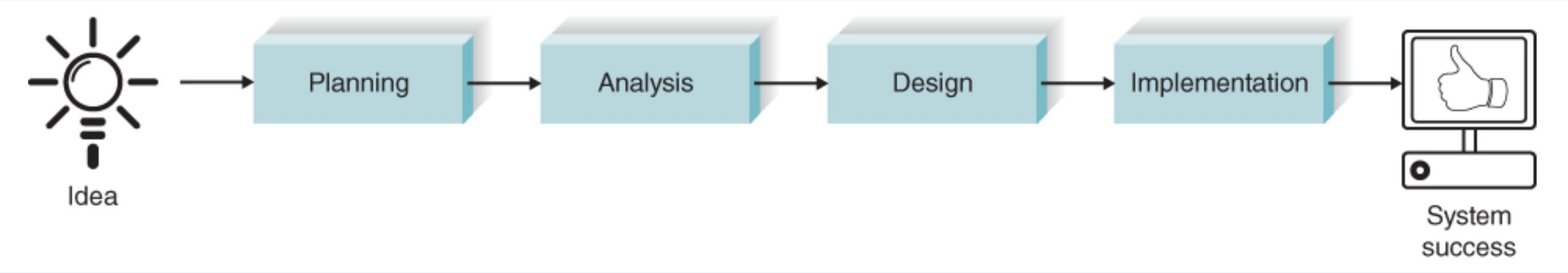
- Infrastructure analyst: technical issues (hardware, software, networks, databases) – new IS conforms to organizational standards, identify infrastructure changes

Systems Analyst Roles (cont.)

- Change management analyst: people and management issues surrounding system installation – adequate documents, support, training
- Project manager: project completed on time and within budget

System Development Life Cycle (SDLC)

- Four fundamental phases
- Each phase composed of a series of steps, which rely on techniques that produce deliverables.
- Different projects emphasise different parts or approach the phases in different ways (e.g. consecutively, incrementally, iteratively)



Planning

- Why IS should be built and how the project team will go about building it
- Two steps:

1. Project initiation
    - Identify system's business value to organization
    - System request – brief summary of business need and how a system will create business value
    - Feasibility analysis

Planning (cont.)

- Feasibility analysis
  - Technical (Can we build it?)
  - Economical (Will it provide business value?)
  - Organizational (Will it be used?)
- Get approval

Planning (cont.)

2. Project management
   - Create a workplan, staff the project, plan how to control and direct the project
   - Project plan – how the team will go about developing the system

Analysis

- Who will use the system, what the system will do, where and when it will be used
- System proposal describing the requirements new system should satisfy
- Three steps:

1. Develop an analysis strategy – studying the as-is system and its problems, envisioning the to-be system

Analysis (cont.)

2. Gather requirements – develop a concept: requirement statements, analysis models (user/system interaction, data and processes needed to support the underlying business process)

3. Create a system proposal presented to decision makers

Design

- How the system will exactly operate in terms of hardware, software, network infrastructure; user interfaces, forms, reports; specific programs, databases, files -> system specification
- Four steps:

1. Design strategy – how the system will be developed (inhouse, outsource, buy)

Design (cont.)

2. Basic architecture design (hardware, software, network infrastructure); interface design, forms, reports
3. Database and file specifications – what data and where data will be stored
4. Program design – programs that need to be written and what each program will do

Implementation

- System is actually built
- Three steps:

1. Construction – build and test
2. Installation – conversion from old to new, training
3. Support plan – systematic way for identifying major and minor changes needed for the system